

# BUSS386 Problem Set 1 — Solutions

## Introduction to Derivatives

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### Problem 1 — Classification

- (i) **Forward.** Korean Air is *long*, the fuel supplier is *short*. *OTC* — bespoke size (10M bbl) and counterparty-specific terms; not a standardized exchange product.
- (ii) **Forward.** POSCO is *short* (delivers steel, receives fixed price); the auto plant is *long*. *OTC* — physical commercial delivery contract, not a standardized exchange product.
- (iii) **Swap** (interest-rate swap). Samsung is the *fixed-rate payer* (long the floating leg); the counterparty is the fixed-rate receiver. *OTC* (vanilla swaps are mandated to clear through a CCP post-2008, but the trade itself is OTC-originated).
- (iv) **Futures.** KRX-listed and marked-to-market  $\Rightarrow$  futures. Long = buyer of the contract, short = seller. *Exchange* (KRX, by construction).
- (v) **Option** (a call). The buyer is *long* the call; the writer is *short*. *Exchange* — standardized strike (₩200,000) and a 3-month tenor are typical of listed equity options.

### Problem 2 — Markets, notional vs. market value

- (a) Notional = \$100M. Gross market value = \$1.2M. Market value better reflects current economic exposure: it is what one party would have to pay the other to tear up the contract today; the \$100M never changes hands.
- (b) Notional measures the face amount each contract is *written on*; market value measures the present value of the *net* expected cash flows. Most contracts trade close to fair value, so the net expected cash flow is small relative to face. Aggregating millions of such contracts gives a huge gap between notional and market value (about 35:1 in BIS data).
- (c)
  - (1) **CCP.** Standardized contract specifications and high volumes make central clearing efficient; KRX clears it.
  - (2) **Bilateral / ISDA.** Deal-specific (tied to one M&A event), so the contract terms cannot be standardized for a CCP.
  - (3) **CCP.** Vanilla swaps between large financial institutions are subject to mandatory central clearing under post-2008 rules (Dodd-Frank in the US, EMIR in the EU).

### Problem 3 — Returns and volatility

- (a) Simple returns:  $R_1 = 2.000\%$ ,  $R_2 = -3.922\%$ ,  $R_3 = 5.102\%$ ,  $R_4 = -1.942\%$ .

- (b) Log returns:  $r_1 = 1.980\%$ ,  $r_2 = -4.001\%$ ,  $r_3 = 4.979\%$ ,  $r_4 = -1.961\%$ .
- (c) Cumulative Day 0 to Day 4:
- Simple:  $\frac{P_4}{P_0} - 1 = \frac{101}{100} - 1 = 1.000\%$  (or, by compounding,  $(1 + R_1)(1 + R_2)(1 + R_3)(1 + R_4) - 1 = 1.000\%$ ).
  - Log:  $r_1 + r_2 + r_3 + r_4 = 0.997\% \approx \ln(101/100) = 0.995\%$ .
- Log returns are easier over time because they *add*; simple returns must be *compounded*.
- (d) Mean of log returns  $\bar{r} = 0.249\%$ . Sample variance  $= \frac{1}{n-1} \sum (r_t - \bar{r})^2 = 0.001610$ . Daily  $\sigma = \sqrt{0.001610} = 4.013\%$ . Annualized  $\sigma_{\text{ann}} = 4.013\% \times \sqrt{252} \approx 63.7\%$ . (High because  $n = 4$  with a large swing; only illustrative.)
- (e) The  $\sqrt{T}$  rule requires daily returns to be **i.i.d.** (independent and identically distributed). Volatility clustering (autocorrelation in squared returns) violates this in practice.

### Problem 4 — Discrete RV (KOSPI 200 scenario)

- (a)  $E(R) = 0.25(0.22) + 0.55(0.07) + 0.20(-0.18) = 0.0575 = \boxed{5.75\%}$ .
- (b)  $\text{Var}(R) = 0.25(0.1625)^2 + 0.55(0.0125)^2 + 0.20(-0.2375)^2 = 0.017969$ .  
 $\sigma(R) = \sqrt{0.017969} = \boxed{13.40\%}$ .
- (c) Expected wealth  $= 50,000,000 \times 1.0575 = \boxed{52,875,000}$ .  
Worst-case (bear) wealth  $= 50,000,000 \times (1 - 0.18) = \boxed{41,000,000}$ .
- (d)  $E(R_p) = 0.5(0.0575) + 0.5(0.03) = 0.04375 = \boxed{4.375\%}$ .  
 $\text{Var}(R_p) = (0.5)^2(0.1340)^2 + (0.5)^2(0.06)^2 + 2(0.5)(0.5)(-0.2)(0.1340)(0.06) = 0.004588$ .  
 $\sigma(R_p) = \sqrt{0.004588} = \boxed{6.77\%}$ .
- (e) Because  $\rho = -0.2 < 1$ , the two assets do not move in lockstep — the cross-term in the variance formula is *negative* when  $\rho < 0$ , so combining them lowers risk relative to either one alone.

### Problem 5 — VaR and Expected Shortfall

$V = \$2,000,000$ ,  $\sigma_{\text{daily}} = 1.1\% \Rightarrow$  daily  $\sigma$  in dollars  $= \$22,000$ .

- (a) 1-day 95% VaR  $= 1.645 \times \$22,000 = \boxed{\$36,190}$ .
- (b) 1-day 99% VaR  $= 2.326 \times \$22,000 = \boxed{\$51,172}$ .
- (c) 10-day 99% VaR  $= 2.326 \times \$22,000 \times \sqrt{10} = \$51,172 \times 3.162 = \boxed{\$161,808}$ .
- (d) Empirical, last 250 days:
- (1) 95% VaR  $= 0.024 \times \$2,000,000 = \boxed{\$48,000}$ .
  - (2) 95% ES  $= 0.030 \times \$2,000,000 = \boxed{\$60,000}$ .

Observe that empirical VaR (\$48k) exceeds parametric VaR (\$36k): the recent SPY tail has been fatter than a Gaussian with  $\sigma = 1.1\%$  would predict.

- (e)
  - (i) ES is the *average* of all returns at or worse than the VaR threshold; since each of those returns is at least as negative as the VaR cutoff, the average is at least as bad as VaR.
  - (ii) In a long calm period (e.g., a 250-day window with no crash), the empirical tail is shallow, so empirical VaR understates the true tail risk that a future crisis would reveal.
  - (iii) Examples: exchange margin requirements (KRX/CME initial margin), insurance solvency capital (Solvency II / K-ICS at 99.5% one-year VaR), asset-manager internal risk limits, corporate Cash-Flow-at-Risk for FX/commodity exposure.